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RESEARCH ARTICLE

Use of superheroes social skills with middle school-age students with autism spectrum disorder

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Abstract

The current study evaluated use of the Superheroes Social Skills program as a means of increasing social skill accuracy in adolescents with autism spectrum disorder. Participants included four Caucasian male students that were eligible for special education services within the autism category. Social skills training was presented twice weekly for 9 weeks. The results demonstrated that implementation of the intervention improved social skill accuracy within the training setting as indicated by visual analysis and nonoverlap of all pairs. Additionally, improvements in social skill accuracy during probes conducted within the classroom with typically developing peers was observed. Despite improvements in social skill accuracy in both settings, sociometric status of participants demonstrated little change from baseline to postintervention.

KEYWORDS

autism spectrum disorder, social skills

Autism spectrum disorder (ASD) affects approximately 1 in 68 children within the population (Centers for Disease Control and Prevention, 2014). One of the hallmark characteristics of ASD includes deficits in social communication and interactions (American Psychiatric Association, 2013). These deficits can have lasting effects on developing and maintaining relationships (Bauminger, Shulman, & Agam, 2003). To remediate and address deficits in social communication and interactions within school settings, social skills training is often recommended and included within students' individualized education plans (IEPs) (Ruble, McGrew, Dalrymple, & Jung, 2010).

Social skills training for individuals with ASD may include a wide range of intervention strategies. These strategies include pivotal response training, environmental modifications, social narratives, adult prompting, and peer-mediated instruction and intervention (Camargo et al., 2014; Rogers, 2000). Additionally, video modeling, self-monitoring, and incorporating typically developing peers in training are evidence-based strategies used within social skills training (Bellini & Akullian, 2007, Davis, Mason, Davis, Mason, & Crutchfield, 2016; Watkins et al., 2015). However, researchers have found a lack of implementation of evidenced-based interventions for children with ASD within school settings (Hess, Morrier, Heflin, & Ivey, 2008; Morrier, Hess, & Heflin, 2011). This could be caused

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by a multitude of reasons, including limited time and resources, lack of support from administration, and minimal exposure to evidenced-based procedures (Collier-Meek, Fallon, Johnson, Sanetti, & Delcampo, 2011; Shernoff, Kratochwill, & Stoiber, 2003). Incorporation of evidence-based intervention strategies into manualized social skills curricula may represent an approach that facilitates use of empirically supported practices in applied settings (Lord et al., 2005).

In addition to the difficulty of translating many empirically supported interventions to applied settings, many social skills interventions have also been found to result in limited generalization of skills across people and settings. Whereas individuals with ASD may demonstrate improvements in skill use within a training environment, effects are often more limited in naturalistic settings (Bellini, Peters, Benner, & Hopf, 2007; White, Keonig, & Scahill, 2007). However, the lack of generalization is not unique to individuals with ASD, as Gresham, Sugai, and Horner (2001) described a lack of generalized changes following social skills training for students with high incidence disabilities. The lack of generalized effects of social skills training may, in part, be a result of factors such as training in contrived settings (e.g., pull-out social skills groups) or lack of programming for generalization (Bellini et al., 2007; Gresham et al., 2001). Thus, to promote generalization of social skills, it is important that social skills training incorporate natural stimuli into the training environment (e.g., setting, peers), program for participants to contact natural consequences of social skill use, and train diversely and loosely (e.g., Gresham et al., 2001; Radley, Ford, Battaglia, & McHugh, 2014a; Stokes & Osnes, 1989).

An example of a manualized social skills curriculum that aims to incorporate multiple generalization-promoting strategies is the Superheroes Social Skills program (Jenson et al., 2011). The Superheroes Social Skills program integrates several intervention strategies found in previous research to promote acquisition of target social skills, namely, video modeling (Bellini & Akullian, 2007), self-monitoring (Davis et al., 2016), behavioral skills training (Nuernberger, Ringdahl, Vargo, Crumpecker, & Gunnarsson, 2013), and social stories (Acar, Tekin-Iftar, & Yikmis, 2016). In addition to inclusion of strategies known to promote acquisition of discrete social skills, the Superheroes Social Skills program aims to enhance generalization of target social skills through incorporation of multiple models of target skills via video models and role-play scenarios and inclusion of peers from naturalistic settings who provide natural consequences of social skills use (e.g., attention, access to activities or tangibles, praise), as well as serve as discriminative stimuli for social skills use in nontraining settings (Radley et al., 2014b).

Multiple studies have evaluated the effectiveness of the Superheroes Social Skills program, both in clinic and school settings. For elementary-age children with ASD, participation in the program has been found to result in increased social engagement during school recess periods and improvements in parent and teacher ratings of social functioning (Radley et al., 2014b; Radley, McHugh, Taber, Battaglia, & Ford, 2015a). Research with kindergarten and elementary-age participants with ASD has also documented improvements in skill accuracy in training settings, with generalization documented during probes conducted with adult social skill partners (Radley et al., 2014b, 2015b; Radley, Hanglein, & Arak, 2016). Whereas the program has primarily been used with elementary-age children with ASD, recent researchers have examined the effect of participation in the program with other populations. O'Handley, Ford, Radley, Helbig, and Wimberly (2016) evaluated the efficacy of the program in increasing skill accuracy in both training and generalization settings for students with intellectual disabilities within a high-school setting. Although originally designed for a younger population (Jenson et al., 2011), results of the study indicated the potential use of the program with older students with developmental disabilities, as all participants demonstrated improvements in skill accuracy in both training and generalization settings following implementation of the intervention.

Given the findings of use of the Superheroes Social Skills program with younger children with ASD, additional research evaluating the effect of the intervention with older children and adolescents with ASD is justified. Thus, the purpose of the present study is to extend the previous literature by evaluating the effect of the program on target skill acquisition with middle-school-age participants with ASD. Whereas generalization of skills has been reported following participation in the intervention, assessment of generalized skill accuracy has primarily been limited to probes conducted with researchers or other adults (e.g., Radley et al., 2014b, 2015b, 2016). As such, the current study also

aimed to extend the literature through evaluation of generalized affects on skill accuracy during probes conducted with same-age, typically developing peers.

1 | METHOD

1.1 | Participants and setting

Participants included four junior-high-school adolescent males recommended by administrators for social skills interventions, based on teacher-reported difficulties interacting with their peers. In addition to a special education classification of autism from a school-based multidisciplinary team, participants were also required to have opportunities to interact with peers in a general education classroom. Assent and consent was received for all invited participants prior to participation in the study. All four participants attended the same public middle school in a rural school in the Southeastern United States. Patricipants had no prior exposure to the intervention, nor was any other social skills instruction prescribed by any of the participants' IEPs. Pseudonyms are used throughout. Ben was a male in fifth grade who received speech therapy services but did not receive special education services for academic subjects. Nick was a male in sixth grade, receiving special education services for reading. Additionally, Nick wore a back brace to correct scoliosis. Ben and Nick were enrolled in a remedial science class with typically developing peers, from which generalized skill accuracy and sociometric data were collected. Jason was a male in seventh grade, earning honor-roll grades. He did not receive special education services for academics. Chris was a male in eighth grade receiving special education services in reading and math. Jason and Chris were enrolled in a remedial social studies class with typically developing peers, from which generalized skill accuracy and sociometric data were collected.

All social skills groups took place in an unused classroom containing several desks, chairs, and a whiteboard. Generalization probes were conducted in the remedial class periods previously described. Probes were presented during bellwork or downtime before dismissal. During this time, all students who have finished their daily assignments are allowed to interact and move around the classroom. The classrooms contained desks, chairs, projectors, a teacher's desk, and, of course, several students.

1.2 | Materials

Materials included the Superheroes Social Skills program (Jenson et al., 2011), a laptop, and board game materials. For the current study, materials used from the program included the treatment manual and program DVDs. DVDs included videos of animated superhero characters who described skill steps and rationale, video models of unknown, similarly aged peers demonstrating target social skills, and animated social narratives. Videos were viewed on a laptop computer and connected to external speakers. Materials for tic-tac-toe, checkers, and drawing were provided to the students to engage with in between/during probes.

1.3 | Dependent measures

1.3.1 | Skill accuracy, training setting

The percentage of skill steps accurately demonstrated in the training setting was the primary dependent variable. Task analyses were developed for each of the skills taught as part of the current study, with each skill being comprised of four to six steps (Table 1). In the training setting, skill accuracy was assessed via researcher-delivered probes. Each probe was initiated by the researcher delivering a predetermined cue for the target skill (e.g., "Hey, [name]" for *body basics*). Following delivery of the cue, researchers recorded the number of skill steps correctly demonstrated. The number of steps correctly demonstrated was then divided by the total number of steps in the task analysis and multiplied by 100.

TABLE 1 Task analyses of target social skills

Body basics	Participate	Expressing wants and needs
1. Face the person with head and shoulders oriented toward conversation partner	1. Demonstrate close physical proximity to the interaction partner (within 1 meter)	1. Initiate and sustain eye contact with activity partner (3 seconds)
2. Make eye contact within 5 seconds and sustain for a minimum of 3 seconds	2. Initiate and sustain eye contact with activity partner (3 seconds)	2. Signal the person you want to talk to (e.g., raise hand, tap on shoulder, say name)
3. Use appropriate voice with a volume appropriate for the setting	3. Wait for turn without skipping others or interrupting turn-taking progression	3. Wait for the person to respond to your signal
4. Use the right expression by having facial expressions match the conversation	4. Join in without disrupting the progression of activity using an activity-appropriate response	4. Say what is wanted or needed (e.g., "I want…")
5. Relax, evidenced by relaxed shoulders, even-paced speech, normal breathing rate, and not playing with objects		5. Wait for the person to respond
		6. Acknowledge with a reply (e.g., okay, thank you)

1.3.2 | Skill accuracy, generalized setting

Generalized skill accuracy served as a secondary dependent variable. Generalized skill accuracy was assessed through peer-delivered probes, with accuracy assessed using the same task analyses (Table 1). All probes were initiated by peer delivery of a cue for skill use, with cues for skill use being different than those used in researcher-delivered probes (e.g., "How are you today" for *body basics*). Following peer delivery of the probe, researchers recorded the number of skill steps correctly demonstrated. The number of steps demonstrated was then divided by the number of total steps in the task analysis and multiplied by 100.

1.3.3 | Social network salience

In addition to probes of skill accuracy in the generalized setting, sociometric data were gathered as an additional measure of generalization. Social network salience (SNS) was assessed through administration of the Friendship Survey (Kasari, Rotheram-Fuller, Locke, & Gulsrud, 2012). During baseline and immediately following termination of intervention procedures, a researcher administered the Friendship Survey to participants and their classmates (i.e., remedial science class for Ben and Nick, remedial social studies class for Jason and Chris). The Friendship Survey contained two questions: "Are there any kids in your class that you like to hang out with," and "Are there kids in your class who like to hang out together?" For the second question, the survey contained instructions to put a circle around each group of children that hung out together. Using procedures developed by Cairns and Cairns (1994), SNS scores were calculated. SNS scores are a representation of the relative social prominence of participants within the classroom. SNS scores are interpreted as isolated (0), peripheral (1), secondary (2), and nuclear (3). Psychometric evaluations have found the Friendship Survey to have high test–retest reliability ($r = .74–.84$), as well as high concurrent validity with direct observation data (Cairns, Leung, Buchanan, & Cairns, 1995).

1.4 | Design

A multiple baseline design across skills, replicated across participants, was used to assess the effects of social skills training on skill accuracy (Cooper, Heron, & Heward, 2007). The study consisted of three phases: baseline, intervention, and maintenance. Phase changes from baseline to intervention and from intervention to maintenance were made dependent upon level, trend, and variability of skill accuracy data from the training setting. Phase changes from baseline to intervention occurred once all participants demonstrated at least two data points indicating stable responding.

Phase changes from intervention to maintenance were made once all participants had demonstrated at least two consecutive data points of 100%.

1.5 | Procedures

Prior to collection of baseline data, teachers of participants were interviewed and asked questions about how well each participant performed the skills taught as part of the Superheroes Social Skills program. Based on interview results, three skills were identified as common deficits across all four participants: *body basics*, *participate*, and *expressing wants and needs*. The *body basics* lesson targeted appropriate nonverbal behaviors such as making eye contact and facing the speaker. The *participate* lesson taught participants to join in groups without disrupting the group's activity. Finally, the *expressing wants and needs* lesson taught students how to appropriately ask for what they might want.

1.5.1 | Baseline

Following the identification of target skills, baseline probes of skill accuracy were conducted. During baseline probes, researchers and peers presented participants with cues for skill use. No performance feedback was provided to participants for either accurate or inaccurate demonstration of the target skill. For the first skill trained, baseline probes of skill accuracy were collected in five sessions.

1.5.2 | Intervention

Intervention sessions were 20–30 minutes in duration, conducted twice per week for 9 weeks. All four participants participated in the same social skills group. All social skills lessons followed the same format. At the beginning of the group, the researcher first reviewed the group rules and purpose of the session with participants. Next, the target skill was introduced by animated superheroes via DVD. Additionally, the animated superheroes provided a rationale for skill use and described discrete steps for performing the skill. Next, participants viewed video models of the target skill. Each video model depicted the same steps as the task analyses used to evaluate skill accuracy. During each session, three to four video models of accurate skill use were viewed. The researcher then provided an in vivo model of inaccurate feedback, encouraging participants to identify skill steps incorrectly demonstrated. Accurate skill use was then modeled by the researcher. Participants then engaged in role-play of the target skill with other participants. During role-play, the researcher provided participants with performance feedback regarding skill accuracy. Role-play was repeated until participants demonstrated 100% accuracy independently. Following role-play, participants viewed an animated social narrative that reviewed skill steps and provided an example scenario of skill use.

Following viewing of the animated social narrative, probes of skill accuracy were conducted. Skill accuracy for those skills in baseline or maintenance was probed first, followed by probes of the skill in intervention. Whereas no performance feedback was provided for skills in baseline or maintenance, praise was provided for accurate demonstration, and error correction provided for inaccurate demonstration of the skill in intervention. Following probes of skill accuracy, participants were thanked for their participation and returned to class.

1.5.3 | Maintenance

After approximately 1 week following demonstration of skill mastery, skills were assessed to determine the accuracy with which skills were maintained. Probes of skill maintenance were conducted using the same probes as during baseline and intervention. No performance feedback was provided by researchers during maintenance phase probes.

1.5.4 | Generalized skill accuracy probes

Generalization probes were conducted in the students' classroom during 10-minute periods of bellwork before instruction or free time before dismissal. Several typically developing, same-age peers, nominated by the teacher, were coached to present the same probes used in training settings for skill use to the participants. Peers delivering cues were not informed that the participants were receiving social skills training. During generalized skill accuracy probes,

peers of participants did not provide performance feedback. Data regarding skill accuracy was collected by researchers using the same task analyses used in training setting probes.

1.6 | Interobserver agreement

Secondary observers independently recorded skill accuracy data to determine interobserver agreement (IOA). IOA was calculated by dividing the number of agreements for skill steps demonstrated divided by the number of agreements and disagreements and multiplied by 100. IOA was calculated for a minimum of 20% of probes per phase. For Ben, IOA was collected during 38% of probes with a mean IOA of 98%. For Nick, IOA was collected during 40% of probes with a mean IOA of 98%. For Jason, IOA was collected during 42% of probes with a mean IOA of 95%. For Chris, IOA was collected during 44% of probes with a mean IOA of 98%.

1.7 | Treatment fidelity

Researchers completed checklists as a measure of treatment fidelity. Items on the checklist included 14 possible steps: collecting baseline data, collecting maintenance data, having all necessary materials, reviewing group rules, review purpose of session/last session, providing a rationale for the skill, playing each of the three videos included in the intervention package, modeling inaccurate skill use, modeling accurate skill use, role-playing with the students, providing feedback based on role-plays, and collecting intervention, maintenance, and generalization data. Steps were marked as *completed*, *not completed*, or *not applicable* (i.e., collecting maintenance is not applicable until a skill has moved from the intervention phase). Treatment fidelity was collected during 100% of sessions with a mean of 100% steps completed. A secondary researcher independently completed the same checklist for 100% of sessions with a mean of 100% agreement.

1.8 | Data analysis

Intervention effect was primarily evaluated through visual analysis of level, trend, variability, overlap across phases, immediacy of effect, and consistency of effect across similar phases (Kratochwill et al., 2010). Additionally, nonoverlap of all pairs (NAP) (Parker & Vannest, 2009) was used to quantify intervention effect. NAP is a nonparametric effect size calculation in which each data point in one phase is compared to each data point in a comparison phase. NAP is interpreted using guidelines provided by Parker and Vannest (2009); scores from .00 to .65 are considered weak effects, scores from .66 to .92 are considered moderate effects, and .93 to 1.00 are considered strong effects. Descriptive statistics were used to evaluate SNS data.

2 | RESULTS

2.1 | Skill accuracy in training and generalization settings

2.1.1 | Ben

Ben demonstrated variable and low-to-moderate levels of skill accuracy for *body basics* during training setting probes, with low levels of skill accuracy demonstrated for all other probes across skills (Figure 1). Following introduction of intervention, immediate improvements in level of skill accuracy were observed. Whereas some variability in skill accuracy was observed for *body basics* and *expressing wants and needs*, levels were generally high. For the *participate* skill, however, initial increases in skill accuracy were not maintained. Following several probes during which low skill accuracy was observed, accuracy increased and was maintained at 100%. Maintenance data demonstrated high levels of skill accuracy for *body basics* throughout the entire phase. Whereas the initial maintenance phase data points for the *participate* skill indicated high skill accuracy, the final data point was 0%. NAP calculations indicated moderate effects for the *participate* skill in both intervention and maintenance for training setting probes, as well as maintenance for

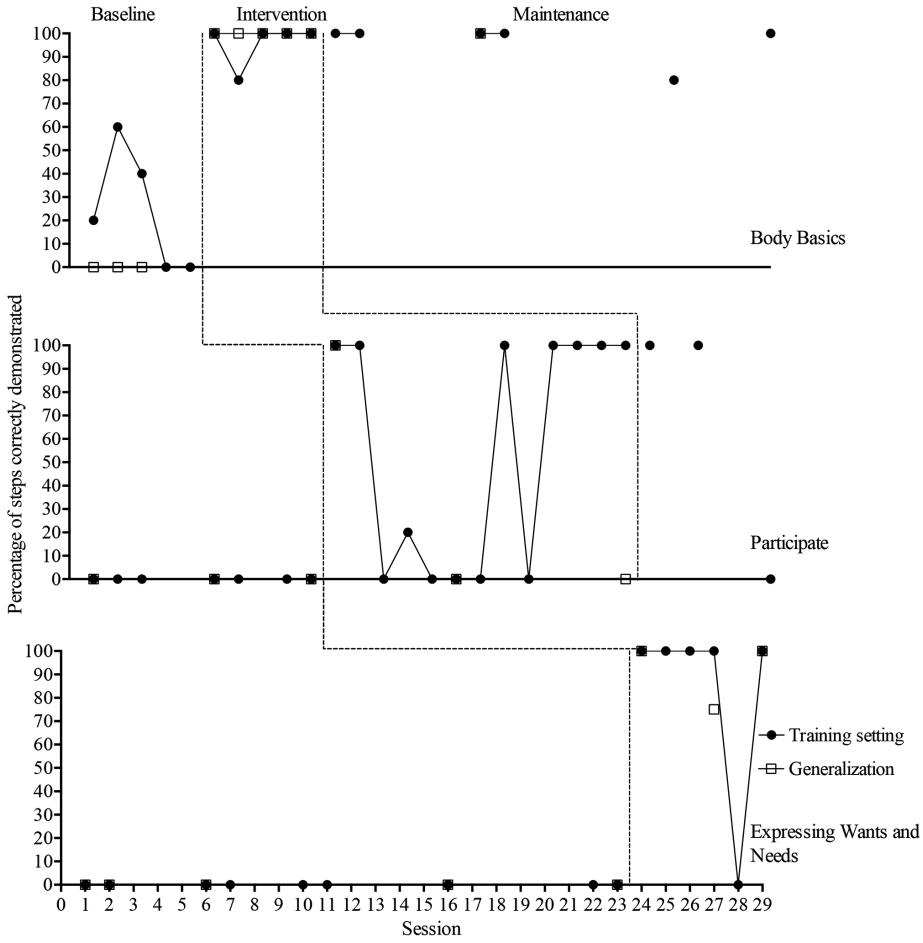


FIGURE 1 Percentage of skill steps correctly demonstrated in probes with researcher and generalization settings, Ben

generalization setting probes. NAP scores for training setting probes of *expressing wants and needs* during maintenance are also interpreted as moderate effects. All other NAP scores are interpreted as strong effects.

2.1.2 | Nick

Nick demonstrated low-to-moderate levels of skill accuracy during baseline probes in both training and generalization settings (Figure 2). Introduction of intervention resulted in immediate increases in skill accuracy across all skills. Whereas high levels of accuracy with some variability were observed for *body basics*, skill accuracy for the *participate* and *expressing wants and needs* skills were 100% throughout intervention. High levels of skill accuracy were observed during the maintenance phase for the *body basics* and *participate* skills. However, decreases in skill accuracy were observed on the final data point for both skills. Anecdotally, Nick was generally noncompliant and refused to participate in data-collection procedures on the final day of maintenance for *participate*. NAP scores for *body basics* for generalization probes during maintenance and *participate* for training setting probes during the maintenance phase are interpreted as moderate effects (Table 2). All NAP scores are interpreted as strong effects.

2.1.3 | Jason

For *body basics*, Jason initially demonstrated low levels of skill accuracy, which increased before stabilizing at 40% (Figure 3). Probes in both training and generalization settings for the *participate* skill remained at 0%. Although

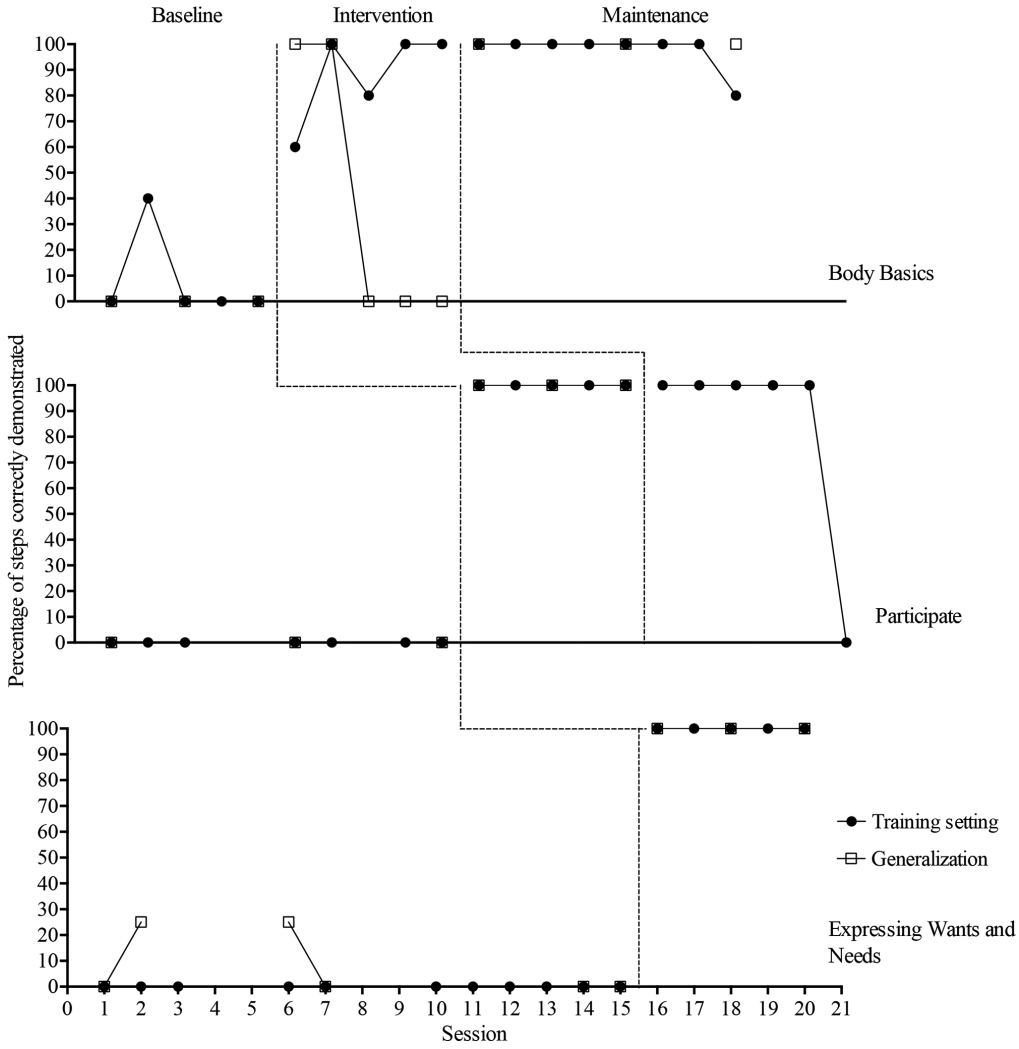


FIGURE 2 Percentage of skill steps correctly demonstrated in probes with researcher and generalization settings, Nick

generalization setting probes for *expressing wants and needs* were initially moderate, a decreasing trend toward 0% was observed. Similarly, all training setting probes for *expressing wants and needs* were at 0%. Following introduction of intervention, skill accuracy for all skills increased to and remained at 100% throughout the intervention phase. With the exception of one data point, skill accuracy remained at 100% for *body basics* during maintenance. Skill accuracy for the *participate* skill was highly variable, initially demonstrating high-skill accuracy before decreasing to low levels, followed by a return to a high level of skill accuracy. With the exception of NAP scores for skill accuracy probes in the training setting for *participate* during the maintenance phase, all other scores are interpreted as strong effects.

2.1.4 | Chris

During baseline, Chris demonstrated low-to-moderate levels of skill accuracy in both training and generalization settings for *body basics*, *participate*, and *expressing wants and needs* (Figure 4). Following introduction of intervention, skill accuracy immediately increased to 100% for all skills in training and generalization settings. Maintenance data collected for *body basics* indicated that skill accuracy was maintained at 100%. Increased variability was observed in skill accuracy for the *participate* skill in comparison to intervention. However, levels of skill accuracy during maintenance

TABLE 2 Nonoverlap of all pairs effect size calculations

	Skill Accuracy			Generalized Skill Accuracy		
	Body Basics	Participate	Expressing Wants and Needs	Body Basics	Participate	Expressing Wants and Needs
Ben						
Intervention	**1.00**	.81	.92	**1.00**	.67	**1.00**
Maintenance	**1.00**	.83	N/A	**1.00**	N/A	N/A
Nick						
Intervention	**1.00**	**1.00**	**1.00**	.70	**1.00**	**1.00**
Maintenance	**1.00**	.92	N/A	**1.00**	N/A	N/A
Jason						
Intervention	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
Maintenance	**1.00**	.75	N/A	N/A	N/A	N/A
Chris						
Intervention	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
Maintenance	**1.00**	.92	N/A	N/A	N/A	N/A

Note. NAP effect sizes scores below .66 are considered small, scores between .66 and .92 are considered moderate, and scores ranging from .93 to 1.00 are considered strong (Parker & Vannest, 2009). Strong effects are in **bold**

represented substantial improvements over baseline levels. With the exception of probes of the *participate* skill in the training setting during maintenance, all other NAP scores are interpreted as strong effects (Table 2).

2.2 | Social network salience

SNS at pre- and postintervention was calculated for all participants. Prior to intervention, Chris, Nick, and Ben received an SNS score of 0. Following intervention, SNS scores for Chris, Nick, and Ben all remained at 0. For Jason, preintervention SNS score was 0. Following intervention, Jason's SNS score increased to 2. Prior to intervention, mean SNS for classroom peers of participants was 1.71. At the conclusion of the intervention period, mean SNS for classroom peers of participants was 1.69.

3 | DISCUSSION

Previous research has found participation in the Superheroes Social Skills program to result in increased skill accuracy in children with ASD (e.g., Radley et al., 2015b, 2016). However, research has primarily included elementary-age participants with ASD. As such, the purpose of the current study is to evaluate the use of the curriculum for middle-school-age students with ASD. Further, the study sought to extend previous research through assessment of skill generalization to novel persons and settings within the school. Results of the study indicate that middle-school-age participants benefitted from participation in the program with improvements in skill accuracy being observed in training settings.

Perhaps more importantly, accurate demonstration of target skills was observed in novel settings with novel persons. Except for the *participate* skill for Ben and *body basics* for Nick, generalization data were found to be high and stable throughout the intervention phase. Whereas generalization has been demonstrated to parents of elementary-age children with ASD (e.g., Radley et al., 2014b, 2016) and teachers of adolescents with intellectual disabilities (O'Handley et al., 2016), no prior studies have assessed generalization of skills by individuals with ASD to other settings and persons within a school setting. Findings of generalization to other settings and persons within the school are notable, given that school-based social skills training programs for children with ASD have been found to produce limited generalized effects (Bellini & Hopf, 2007). Findings of generalization may be attributable to incorporation of multiple

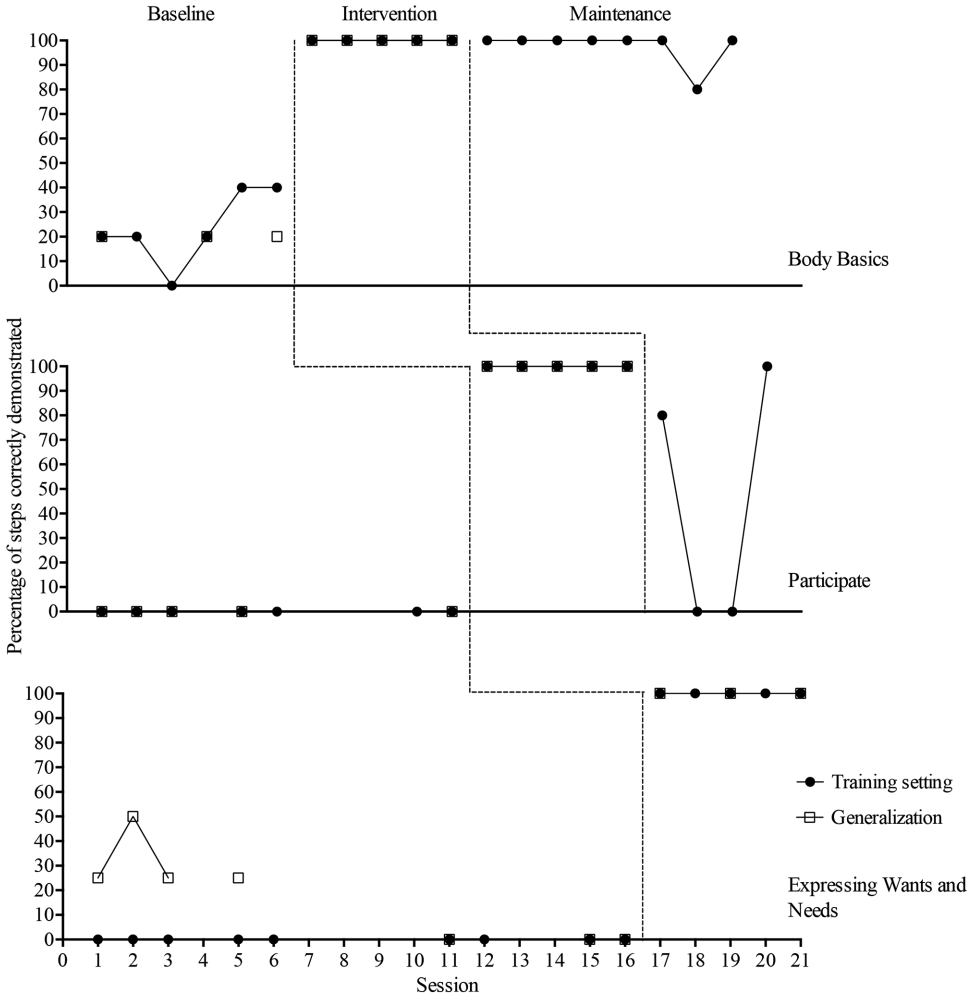


FIGURE 3 Percentage of skill steps correctly demonstrated in probes with researcher and generalization settings, Jason

exemplars of target skills provided to participants (i.e., in vivo models, video models, role plays, social narratives), the presence of naturalistic stimuli within the training settings (i.e., classroom, peers), and the delivery of natural consequences of social skill use via peers.

Whereas the results of the study support previous findings regarding the effect of participation in the Superheroes Social Skills program on accurate demonstration of target social skills, sociometric outcomes in the current study differ from previous research. Although several previous studies found sociometric ratings to increase following participation in the program (Radley et al., 2014b, 2015a), only one participant demonstrated improved sociometric outcomes in the current study. Differences in outcomes may, in part, be explained by challenges associated with sociometric assessment in middle-school settings (Poulin & Dishion, 2008). Primarily, sociometric data were only assessed in one classroom of the target students. Whereas elementary-age participants are typically in the same classroom for the entire day, increasing contact with that peer group, participants in the current study frequently changed classes. As middle-school-age students do not interact exclusively with classmates (Poulin & Dishion, 2008), it may be that the sociometric data collected in the current study did not adequately capture the effects of participation in the intervention on sociometric status. As an alternative hypothesis, the middle-school-age peers may have had long histories with the target students in which social interactions were aversive. Thus, it may be that the relatively brief

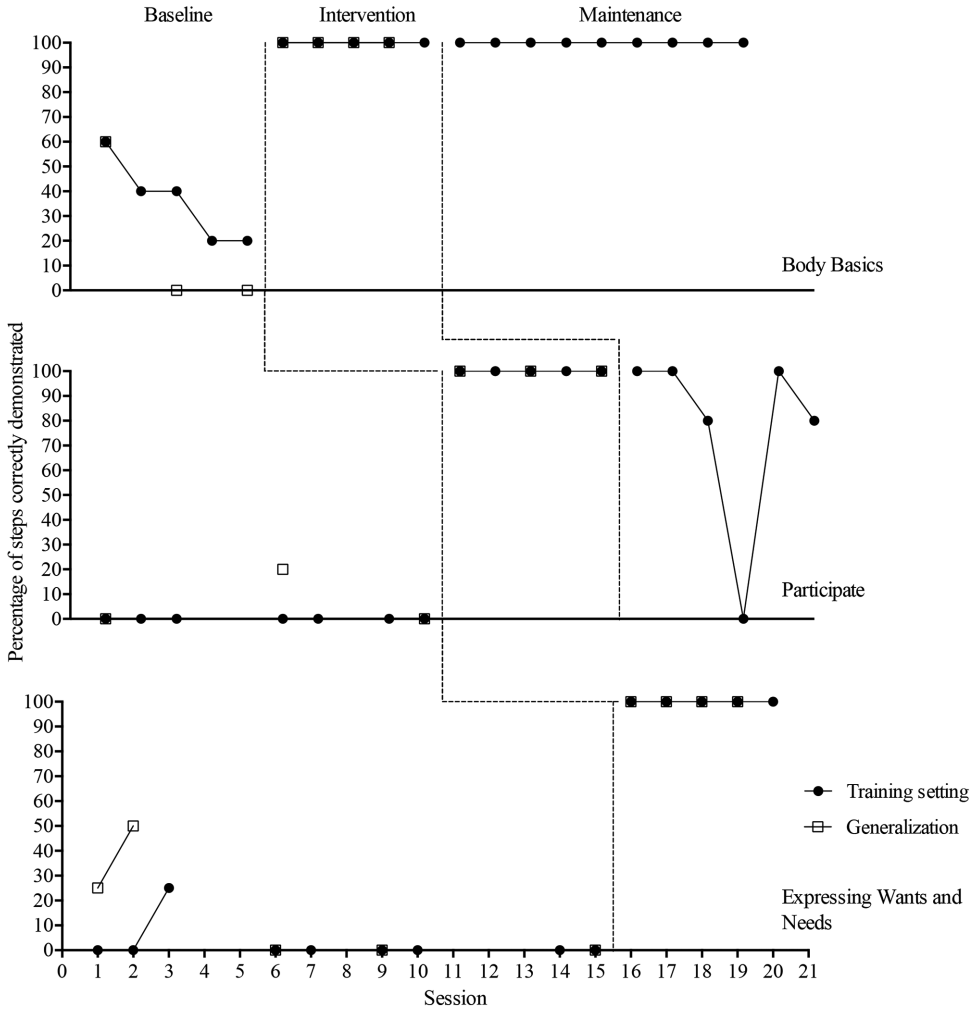


FIGURE 4 Percentage of skill steps correctly demonstrated in probes with researcher and generalization settings, Chris

duration of the intervention was insufficient to overcome well-established learning histories of peers regarding social interaction skills of target students.

3.1 | Limitations

The results of the current study must be interpreted considering several limitations. First, the intervention was facilitated by trained graduate students who served as behavioral consultants to the school and had previous experience in implementation of social skills interventions. Although facilitation of such social skills groups was within the scope of their duties as behavioral consultants, it is unknown whether similar results would be observed if the program were facilitated by untrained school personnel. Thus, it is critical that future evaluations of the program consider having instruction facilitated by school-based staff (e.g., special education teachers). Second, no maintenance data were collected for the *expressing wants and needs* skill for any of the participants because of time constraints associated with the end of the school year. As no maintenance data were collected, findings regarding the maintained effects of the Superheroes Social Skills program for middle-school-age individuals with ASD should be considered preliminary and in need of further evaluation. Third, no generalization data were collected during maintenance because of time constraints.

Thus, it is unknown whether improvements noted during generalization probes were continued following termination of intervention. Fourth, generalization probes were conducted with familiar individuals (i.e., peers) in a familiar setting. Although findings of generalization of skills from the training setting to this setting are important, future researchers should also consider assessment of generalization to unknown persons or novel settings. Future researchers may also consider strategies for assessment of generalized skill use that do not require a researcher to be present, which may result in participant reactivity. Finally, the current study only assessed sociometric outcomes of participants in one setting. As previously discussed, this may have been insufficient for documenting changes across all settings within the school. In the future, researchers may consider conducting sociometric surveys in all classes of middle-school-age participants with ASD.

4 | CONCLUSION

Previous researchers have found participation in the Superheroes Social Skills program to result in improved skill accuracy in both training and generalization settings (Radley et al., 2014b, 2015b). These findings diverge from literature finding pull-out social skills program to result in limited generalization (Bellini et al., 2007). The current study adds to the literature with regard to the use of the Superheroes Social Skills program, indicating that the program may also be useful in promoting skill accuracy and generalization in middle-school-age individuals with ASD.

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